GABRYEL VERISSIMO

Software Engineer · Fintech & reliable systems

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PROFILE

Software engineer building production-grade services in Python (FastAPI) and Java (Spring Boot), with a focus on fintech APIs — payments, ledgers, idempotent transfers, and secure authentication — and shipping fullstack tools with React and TypeScript. Comfortable with PostgreSQL, REST design, Docker, and test-driven development. Seeking graduate and junior software engineering roles in London from summer 2027.

TECHNICAL SKILLS

Languages: Python, Java, C#, SQL, JavaScript, C++, HTML/CSS

Backend: FastAPI, Spring Boot, Spring Security, REST APIs, PostgreSQL, Redis, SQLAlchemy, Alembic, JWT authentication

Frontend: React, Next.js, TypeScript, Vite, Tailwind CSS, REST API integration

Tools & Practices: Docker, Git, GitHub Actions (CI), pytest, Vitest, test-driven development, background workers & queues, Linux, AI-assisted development (Claude, GitHub Copilot)

PROJECTS

PayLedger — Double-Entry Payments API

Python · FastAPI · PostgreSQL · Alembic · Docker | github.com/gabryelvs/payledger | live demo

- Double-entry payments API: users hold multi-currency wallets and transfer money, with every movement recorded as balanced, append-only ledger entries; only the owner can move or read a wallet's money.
- Transfers are race-safe (row-level locks in ascending id order, balance checked under the lock) and idempotent (retry key claimed in the same transaction, scoped per user, so retries can't double-charge).
- Test-driven (67 pytest tests against real PostgreSQL, incl. concurrency tests), with Docker Compose, GitHub Actions CI, and Alembic migrations.

Webhook Inspector — Fullstack Webhook Debugging Tool

Python · FastAPI · PostgreSQL · React · TypeScript · Tailwind · Docker | github.com/gabryelvs/webhook-inspector | live demo

- Fullstack tool for debugging webhooks: users create a disposable URL, point any provider at it, and watch requests arrive live in a React interface showing headers, pretty-printed body, and query parameters.
- Hardened for public deployment: per-client rate limiting keyed on the real client IP behind the proxy, streamed request bodies capped at 1 MB so a large payload cannot exhaust memory, per-bin retention limits, and a capture endpoint that always answers 200 so a database fault never breaks the sender's webhook.
- Test-driven across the stack (47 backend pytest + 7 frontend Vitest tests), the React app built and served alongside the API, deployed on Vercel with Neon PostgreSQL.

FX-Service — Async Currency Exchange API

Python · FastAPI · httpx · Redis | github.com/gabryelvs/fx-service | live demo

- Async microservice serving live currency rates and conversions, pulling ECB rates and caching them in Redis with an in-process scheduled refresh loop.
- Resilient by design: continues serving last-known rates when the upstream provider is unavailable (stale fallback), proven by an automated outage test.
- Test-driven (31 tests, 90% coverage) with Docker Compose and GitHub Actions CI.

Webhook-Dispatcher — Reliable Webhook Delivery

Python · FastAPI · Redis | github.com/gabryelvs/webhook-dispatcher

- Service that reliably delivers webhooks via a Redis queue and a separate worker process, signing each request (HMAC-SHA256) so receivers can verify authenticity.
- Retries failed deliveries with exponential backoff and jitter, dead-letters exhausted ones, and supports manual replay; at-least-once delivery with de-duplication headers.
- Test-driven (28 tests, 92% coverage) with Docker Compose and GitHub Actions CI.

Taskboard API — Trello-like Task Manager API

Java 21 · Spring Boot · PostgreSQL · Docker | github.com/gabryelvs/taskboard-api | live demo

- Trello-like task manager REST API with JWT authentication using refresh-token rotation (reuse revokes every session for that user), and role-based project membership (OWNER/MEMBER) with 404-no-leak authorization.

- Transactional drag-and-drop card ordering with pessimistic column locking (deterministic lock order) proven under concurrent-move integration tests; RFC 7807 problem+json error responses.
- Test-driven (72 tests: 59 Testcontainers integration tests against real PostgreSQL, 13 unit), OpenAPI/Swagger docs, GitHub Actions CI, and deployed on Render with Neon PostgreSQL.

SECTOR—9 — Animated Demo Storefront

TypeScript · Next.js 16 · React 19 · Tailwind 4 · GSAP · Cloudflare Pages | github.com/gabryelvs/store-demo | live demo

- Storefront demo built as a client-facing sales asset: 20-product catalogue, hover-swap product grid, quick-view size selection and a persistent cart drawer, with the catalogue behind a single data seam so a real backend replaces it without touching the pages.
- Scroll-driven GSAP reveals and a parallax band, with three overlays (quick-view, cart, mobile navigation) that each trap focus, mark background content inert for screen readers, restore focus to their opener, and stand down entirely under prefers-reduced-motion.
- Cart is a pure reducer with derived totals in integer pence and validated localStorage rehydration (30 Vitest tests); 26 statically prerendered routes served as plain files from Cloudflare Pages, scoring 100 on Lighthouse (desktop) for performance, accessibility, best practices and SEO, with zero layout shift.

OWASP Security Lab — Break it & Fix it

Python · FastAPI · pytest · Docker | github.com/gabryelvs/owasp-security-lab

- Intentionally-vulnerable FastAPI app demonstrating six OWASP Top 10 issues (SQL injection, broken access control, SSRF, JWT authentication flaws, sensitive-data exposure, and security misconfiguration), each with a working exploit, a hardened fix, and tests proving both.
- Test-driven (14 exploit/fix tests) with GitHub Actions CI; documents the attack, root cause, and remediation per vulnerability — demonstrating secure-coding and defensive engineering.

WORK EXPERIENCE

Greenwich Internship – Consultancy Sprint, University of Greenwich — Oct 2026 – Nov 2026

- Working in a six-person team on a live brief from an external organisation assigned by the University: researching the client's challenge, analysing findings and developing evidence-based recommendations for a consultancy-style report, with feedback from the client.

Software Engineering Intern, AP Homes Ltd, London — Sep 2026 – Oct 2026

- Designed, built and deployed the production website for Auto Boutique London, a Central London car-storage business, replacing its WordPress site with a static Astro build and a PHP enquiry endpoint that sends mail through Resend.
- Set up push-to-deploy from GitHub Actions to SiteGround over SSH, archived the old WordPress install with verified backups, and completed a UK consumer-law pass: legal pages, consent-gated enquiry form, and no unverified reviews on the site.

Runner, The Ivy Market Grill, Covent Garden — Jun 2026 – Sep 2026

- Front-of-house runner in a restaurant serving 400–600 covers a day: ran food to section, handled allergen and dietary requirements, and kept pace and accuracy through peak service.
- Held accuracy and composure through the busiest services; trained new starters on section layout, allergen handling and service flow.

Youth Leader (Volunteer), Cathedral International, West Norwood — Jun 2023 – Present

- Lead a weekly session for 10–15 teenagers as part of the youth leadership team, covering mentoring, pastoral support, and planning session content.
- Co-organised two youth events for 100–150 attendees, owning the technical side: live projection, visual design and art direction.
- Serve on the church projection team (2024 – Present), operating live visuals across weekly services — cueing lyrics, media and slides in real time with no pre-built running order (ProPresenter).

EDUCATION

BSc Computer Science — University of Greenwich (2023 – expected Jul 2027): Final year, following a foundation year.

City of Westminster College (2022 – 2023): BTEC Level 2 ICT · GCSE English. Lambeth College (2021 – 2022): Level 1 Diploma in ICT · GCSE Maths · Functional Skills Maths & English. SESI-SENAI College (2019 – 2020): Diploma in Microsoft Applications.

LANGUAGES

Portuguese: Native **English:** Fluent (advanced — speaking, reading, and writing)